

RNSC: A hierarchical deep learning model for net promoter scoring understanding by combining review and note through semantic consistency

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ABSTRACT

The Net Promoter Score (NPS) is a widely-used metric for measuring customer loyalty and is an essential element in customer service analysis. To understand and analyze NPS effectively, customer-generated content, such as customer reviews, is commonly employed. Nevertheless, the information provided in customer-generated content may sometimes be inadequate. In this paper, we aim to enhance NPS understanding by incorporating a valuable employee-generated content known as employee notes, which provide insights into the internal operations of the enterprise's customer service. Considering the different standpoints of the two pieces of textual information, the semantic consistency is defined to calculate the semantic relationship between them, which helps better understand and analyze NPS. To effectively capture the crucial semantic consistency relationship and enhance the understanding of NPS, we propose a hierarchical deep learning approach, called RNSC (Review, Note and Semantic Consistency), which employs a bidirectional LSTM (BiLSTM) and a semantic consistency recognition module to encode and identify the local relationship between customer reviews and employee notes. Based on the trained model, the attention mechanism can detect and output the keywords that significantly contribute to the NPS through semantic consistency, showing its potential for supporting diagnostic analysis and further service optimization. Extensive data experiments, such as comparison with baseline methods, ablation experiments, and sparse data experiments, verify the superiority of the proposed RNSC approach.

1. Introduction

The Net Promoter Score (NPS) is a well-established metric used to assess customer loyalty [1]. The responses classify customers into three categories: promoters, passives, and detractors [2,3]. Surely, detractors with low NPS ratings are of the major concern for customer service analysis and improvement, as low NPS ratings indicate a low level of customer loyalty, which can directly lead to customer churn and negative word-of-mouth, ultimately impacting business profitability [4,5]. Therefore, companies strive to explore the reasons behind low NPS ratings in their services and subsequently propose targeted remediation.

Customer review, as a typical form of customer-generated content, usually conveys the voice of the customer by describing and evaluating the service. An emerging stream of research has attempted to understand NPS derived from such versatile information resources about customers' concerns [6,7]. Nevertheless, unspecialized customer reviews suffer from inherent bias or incomplete information regarding the service, as they often deviate from comprehensive content, especially in specialized or technical service contexts, and primarily focus on their emotional

expressions rather than providing detailed and balanced information [8,9]. Furthermore, customers often submit reviews spontaneously and optionally, resulting in a scarcity of data [10]. Considering these limitations, while customer reviews have the potential to understand NPS, it is crucial to gather additional information resources to complement the reviews.

To address the issue of incomplete information in understanding and analyzing NPS, this study incorporates employee notes as an alternative and valuable text source to compensate for the limitations in customer reviews. After a service has been delivered, the customer is prompted to provide an NPS rating and an optional review, while the enterprise requires the employee who delivers the service order to note the related information for internal operational purposes. These records usually contain professional wording, technical details, and key identification cruxes relating to the service, particularly for cases with low NPS ratings. This might effectively compensate for the information incompleteness and data sparsity in customer reviews. Customer reviews and employee notes reflect feedback from customers and service records from employees regarding the same service, corresponding to the

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generation process of a unified NPS. Therefore, the fusion of customer reviews and employee notes may help enrich service-related information and subsequently enhance NPS understanding and analysis [11,12].

Existing research on customer rating analysis mainly focus on unilateral reviews [6,7], neglecting the value of integrated analysis of bilateral texts like customer reviews and employee notes. The research gaps are mainly reflected on two aspects. Firstly, even though some text processing research has explored bilateral information such as question-answer texts, there is still a need for further analysis on how to depict and detect comprehensive relationships for paired information from customers and employees with different perspectives (e.g., inconsistent) on the same service, so as to uncover insights for better understanding NPS. Secondly, building on this, designing effective intelligent methods to model and capture the relationships for enhancing the fitting accuracy of NPS, as well as further providing an explanation at the granularity of keywords for NPS diagnosis, is another research gap to address.

To fill the first gap, we will utilize the findings of relevant theoretical researches and typical cases in practice data to extract and analyze the comprehensive relationship between customer reviews and employee notes. Stakeholder theory [11] proposes that external customer-generated content, such as customer reviews, help detect and address the significant deviations in customer attitude and further match the service practices expected by customers. Comparatively, internal employee-generated content, such as employee notes, contributes to identifying key cruxes, searching for operational solutions, and prioritizing actionable intelligence. Therefore, customer reviews and employee notes can complement each other and effectively contribute to understanding and analyzing NPS. Furthermore, customers and employees may describe the same service differently based on their perspectives. The semantic consistency between reviews and notes can offer additional insights to better understand NPS, an area that has been largely unexplored. Fig. 1 illustrates two real-world service examples from a telecom company to showcase the correlation between semantic consistency and NPS. In the first example where NPS = 10, the customer gives a positive rating, which might seem somewhat inconsistent through direct text mining analysis on customer review. However, considering the bilateral textual relationships, both the customer evaluations and employee records are aligned, indicating a high level of semantic consistency between the review and note. This alignment elucidates why the NPS is high at 10. Conversely, in another example where NPS = 4, explicit semantic inconsistency is evident concerning response efficiency, landline issues, and noise when comparing the review and note. This explains that, even though the customer review is positive (e.g., on response efficiency), the NPS is still low (because the key issues regarding landlines and noise are not solved).

By combining employee notes with customer reviews, the NPS analysis can be greatly enhanced, as the combining can provide complementary and valuable information. A couple of studies examined the

semantic consistency between internal management information and consumer comments, showing its influence on customer evaluation [13]. Related researches show that a topic-matching between internal management information and consumer reviews might increase the perceived effectiveness and reduce potential perceived biases and cognitive cost for peer consumers as well [14,15]. The consistency in the semantics of information generated by customers and employees can reflect the consensus between the service requirements of consumers and the understanding and provision of those requirements by employees. This alignment can mirror the degree to which customer needs are fulfilled and influence customers' subsequent evaluations, that is, the NPS. This highlights that combining customer reviews and employee notes can compensate for the information incompleteness and data sparsity to understand and analyze NPS, which can be enhanced with semantic consistency recognition.

Building upon the semantic consistency relationship, we can devise effective intelligent approaches to address the second gap. This study proposes a deep learning-based NPS understanding and analysis approach, namely RNSC (Review, Note and Semantic Consistency), to investigate NPS by incorporating customer reviews and employee notes that are enhanced using semantic consistency. The RNSC approach embraces a hierarchical deep learning network based on BiLSTM and a semantic consistency recognition component to learn the semantic consistency of customer reviews and employee notes across the word and latent semantic levels. Specifically, we design an element-wise product and subtraction as a semantic consistency recognition module to measure the similarity and difference between paired semantic representations of customer reviews and employee notes. Ultimately, a multi-layer perception (MLP) is applied to evaluate the NPS using these comprehensive representations of reviews and notes. Based on the trained model, the attention and semantic consistency score are calculated on the test set to detect the consistency features and extract texts. This helps automatically identify the keywords that significantly contribute to low semantic consistency and NPS. By conducting extensive experiments on real-world customer service feedback datasets and comparing them with baseline methods, the superiority of the RNSC approach is verified.

This study introduces a novel NPS analysis framework and a deep learning approach that integrates paired texts from customer reviews and employee notes through the lens of semantic consistency to identify cruxes of low NPS, for better NPS diagnosis. The RNSC deep learning approach demonstrates its advantages on two aspects. First, by leveraging information from employee notes together with customer reviews through the lens of semantic consistency, it shows its superiority on the fitting performance of NPS. Second, by integrating effective semantic consistency relationship and attention mechanism, it offers a keyword-level interpretation for better NPS diagnosis.

The remainder of the study is organized as follows. Section 2 briefly reviews the related literature. The technical details of the proposed



Fig. 1. Two real-world service examples from a telecom company.

RNSC approach are introduced in Section 3. In Section 4, empirical studies and data experiments are extensively discussed. Section 5 concludes the study.

2. Related literature

2.1. Machine learning for customer rating learning and analysis

Machine learning-based methods are an important part of customer rating and loyalty learning and analysis [16–19]. For instance, Zaki et al. [3] have extended the linguistic text-mining approach to determine each customer's complaint status and emotions based on customers' data from different sources. They have also used text-mining feedback to divide customers into groups of complainers, neutral, or satisfied [3]. Topic model-based methods have been widely deployed to extract topics from reviews to analyze NPS [20–22]. For instance, CTR combines the merits of traditional collaborative filtering and probabilistic topic modeling, which can be used for in- and out-of-matrix predictions [22]. Cheng et al. [20] have further relaxed the constraint of one-to-one mappings between potential topics and factors. Sengupta et al. [21] have used a probabilistic framework based on latent Dirichlet allocation to obtain subjective information such as opinions, attitude and feelings expressed in short texts and associate sentiments with them.

Recently, deep learning methods have been widely favored because of their robust learning performance [23,24]. The convolutional neural network (CNN) and Recurrent Neural Network (RNN)-based architecture are widely employed to extract the user's latent features from the corresponding reviews for customer ratings or preference learning [24, 25]. To learn the potential influencing factors, Li et al. [14] August] have proposed a deep learning-based framework named NRT, which can simultaneously predict precise ratings and generate abstractive tips. Moreover, several studies have been proposed to discriminate the importance of each latent feature or factor to rating [26,27]. Using the attention mechanism, models can pinpoint crucial keywords from textual auxiliary information, thereby offering semantic explanations for recommendations. [27–30]. For instance, Lu et al. [27] have presented TARMF utilizing attention-based recurrent neural networks to extract topical information from reviews. To capture the aspects of products/services and the underlying sentiments associated with the aspects, Da'u and Salim [31] have proposed a sentiment-aware deep recommender system with neural attention network (SDRA), which introduces a semi-supervised topic model to extract aspects of product/service and sentiment lexicons.

Table 1 compares the above-listed methods on customer rating learning with the proposed RNSC approach in terms of the key designing points, i.e., (1) whether the method understands and analyzes ratings at a fine-grained level, such as analyzing ratings at the aspect level; (2)

whether it utilizes deep learning methods for latent feature representation learning; (3) whether the method addresses the issue of sparse customer reviews by using multiple sources of information; and (4) whether it integrates semantic relationships in processing multi-source data. A key challenge associated with these technical solutions lies in their emphasis on capturing the impact of customer emotion or individual preference on ratings, primarily relying on customer feedback data. However, the quality of this information may be weak and lack of completeness, resulting in reduced accuracy in NPS evaluation and analysis [9]. Therefore, an avenue worth exploring is the integration of customer reviews and employee notes within the neural network framework. Nevertheless, existing studies based on deep learning have overlooked the semantic relationship between these two sources of information, such as the notion of semantic consistency. Hence, it is essential to incorporate effective mechanisms that seamlessly fuse paired information into deep learning models for the analysis and comprehension of NPS, potentially enhancing the learning performance of the model.

2.2. Semantic relationship recognition

As a central problem in natural language understanding, Natural language Inference (NLI) leads to the problem of determining entailment and contradicting relationships between the premise and hypothesis [32,33]. These researches exploring the relationship between paired texts are consistent with our goal of recognizing semantic consistency between customer- and employee-generated content. The recent availability of large-scale annotated datasets has enabled researchers to develop various deep learning-based architectures for NLI, such as the large Stanford Natural Language Inference (SNLI) corpus [34,35] and Multi-Genre Natural Language Inference (MNLI) corpus [36]. Therefore, we utilize large, annotated datasets to train semantic inference models. Next, the trained model is used to learn the semantic consistency between paired customer reviews and employee notes.

Neural networks have become a predominant method used to explore text similarity. The dominating trend in NLI models is to build complex and deep text representation models [37]. Specifically, Munkhdalai et al. [38] have proposed complicated networks that consider sequential LSTM-based encoding, recursive networks, and complicated combinations of attention models. To make the model more trivially parallelizable, Parikh et al. [39] have proposed an attention-based model, which relies on alignment and is fully computationally decomposable to the input text. Ghaeini et al. [40] have proposed a novel reliant reading bidirectional LSTM network (DR-BiLSTM) to efficiently model the relationship between the premise and hypothesis during encoding and inference. NLI-related methods can effectively mine the latent semantics of unstructured texts and learn the

Table 1
Comparison with previous methods.

	Fine-grained understanding of ratings	Deep representation learning	Multiple information sources	Semantic relationships across multiple sources
Wang et al. [22]	✓	×	✓	×
Zaki et al. [3]	×	×	×	×
Li et al. [14]	×	✓	×	×
Chen et al. [26]	×	✓	×	×
Cheng et al. [20]	✓	×	×	×
Lu et al. [27]	✓	✓	×	×
Da'u and Salim [31]	✓	✓	×	×
Sengupta et al. [21]	✓	×	×	×
Bi et al. [28]	✓	✓	×	×
Wassouf et al. [19]	✓	×	✓	×
Hyun et al. [29]	×	✓	✓	×
Li et al. [30]	✓	✓	×	×
Deng et al. [16]	×	✓	×	×
Suh [18]	✓	×	×	×
Our approach (RNSC)	✓	✓	✓	✓

relationships between latent semantics. Therefore, they have the potential to process and detect the semantic consistency between paired texts. In the aforementioned discussion, customer reviews and employee notes offer paired service-related data from external and internal enterprises. There may be semantic consistency, partial consistency, or inconsistency between them. Therefore, the well-established NLI model trained using large, annotated datasets can be utilized to detect any semantic consistency [35]. Thereafter, the degree of correlation between semantic consistency and NPS can be further justified, which provides a basis for constructing a novel NPS understanding and analysis approach that examines customer reviews and employee notes in a sophisticated manner.

3. The RNSC approach

Generally, for a customer service NPS analysis task, given customer reviews, employee notes as well as corresponding NPS ratings, the research goals are, first, modeling and aligning bilateral texts with semantic features; second, learning and recognizing the semantic consistency between reviews and notes with NPS ratings as label; third, detecting the keywords or phrases in reviews and notes with semantic consistency and attention scores to help diagnosis. To achieve the goals, a RNSC approach is proposed, which comprises the following major components: BiLSTM-based encoding layer, attention layer, semantic consistency recognition layer, and Multi-Layer Perception (MLP)

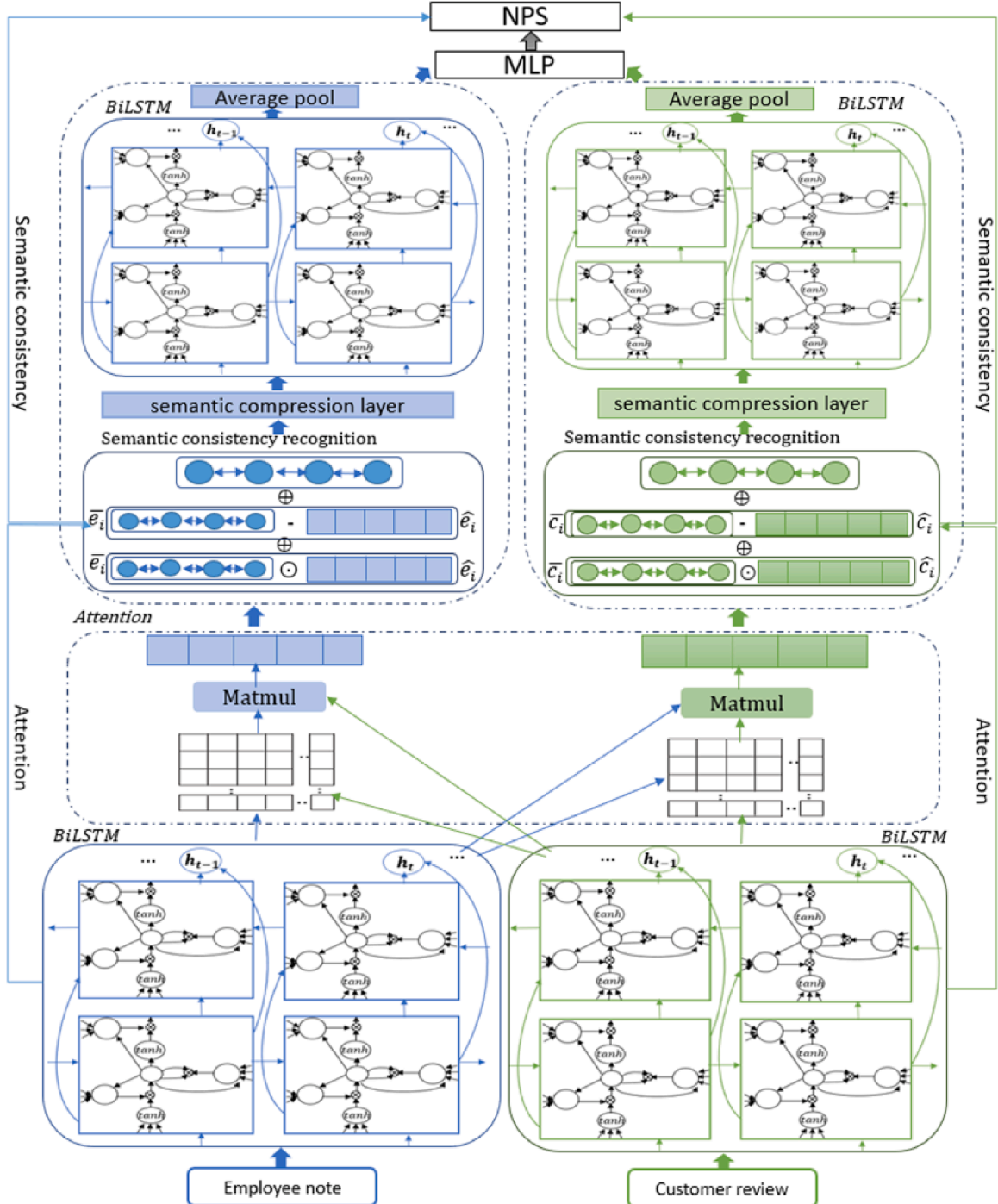


Fig. 2. The framework of the RNSC approach.

learning layer. For a service with an employee note (denoted as e), a customer review (denoted as c) and a NPS rating, $e=(e_1, e_2, \dots, e_{l_e})$ and $c=(c_1, c_2, \dots, c_{l_c})$, where e_i or $c_j \in R^r$ is a word-embedding of an r -dimensional vector (which can be initialized with word2vec), and l_e and l_c are the max length of e and c , respectively. Semantic consistency can be defined as a comprehensive consistency between e and c , which includes the consistency of emotional polarity, topic distribution and other potential semantic features between the two texts, which are denoted as con_{ec} . Thus, with the NPS rating as label, the semantic consistency score of each pair of e and c can be learned with the trained model. In addition, an attention score is further calculated to map the semantic consistency to specific word granularity in e and c to achieve in-depth understanding and analysis of NPS. Fig. 2 presents the framework of the RNSC approach.

3.1. BiLSTM-based encoding

RNNs are widely used in sequence information processing, and LSTM is a variant of RNN designed to deal with the vanishing gradients problem [41]. Therefore, this study utilizes bidirectional LSTM to encode customer reviews and employee notes, which can hold valuable information over long intervals and dispose of useless information [42, 43]. BiLSTM uses two LSTMs to learn each token of the sequence based on the post and next context of the token, where the forward $LSTM^{\rightarrow}$ processes the sequence from left to right and the backward $LSTM^{\leftarrow}$ processes the sequence from right to left. Given a sequence of word representation, the i -th hidden state h_i is encoded as formulas (1–3), where x_i represents the input. The word-embedding sequences from customer reviews and employee notes can be used as input for BiLSTM to get $\bar{e}_i$ and $\bar{c}_j$, as showed in formulas (4, 5).

$$h_i^f = LSTM^{\rightarrow}(x_i, h_{i-1}^f) \quad (1)$$

$$h_i^b = LSTM^{\leftarrow}(x_i, h_{i+1}^b) \quad (2)$$

$$h_i = h_i^f \oplus h_i^b \quad (3)$$

$$\bar{e}_i = BiLSTM(e_i, h_{i-1}), i \in [1, 2, \dots, l_e] \quad (4)$$

$$\bar{c}_j = BiLSTM(c_j, h_{j-1}), j \in [1, 2, \dots, l_c] \quad (5)$$

3.2. Attention layer

Motivated by NLI, which models local inference by employing hard or soft alignment to associate the relevant subcomponents [44], a soft attention mechanism is employed to identify the local relationship between customer reviews and employee notes. As shown in formula (6), the similarity between the hidden states can be depicted through an unnormalized attention weight. Next, the relevant initial semantic representations can be collected according to the s_{ij} , where the $\hat{e}_i$ and $\hat{c}_j$ are the compositive semantic representations generated by the attention layer, as shown in formulas (7, 8). Formula (7) learns the relevant information related to employee notes in customer reviews and uses cross-attention to embed this information into the $\hat{e}_i$. Similarly, $\hat{c}_j$ learns the key information related to customer reviews in employee notes.

$$s_{ij} = \bar{e}_i \bar{c}_j^T, i \in [1, 2, \dots, l_e], j \in [1, 2, \dots, l_c] \quad (6)$$

$$\hat{e}_i = \sum_{j=1}^{l_c} \frac{\exp(s_{ij})}{\sum_{n=1}^{l_c} \exp(s_{in})} \bar{c}_j, i \in [1, 2, \dots, l_e] \quad (7)$$

$$\hat{c}_j = \sum_{i=1}^{l_e} \frac{\exp(s_{ij})}{\sum_{m=1}^{l_e} \exp(s_{mj})} \bar{e}_i, j \in [1, 2, \dots, l_c] \quad (8)$$

3.3. Semantic consistency recognition

To recognize the semantic consistency, the degree of similarity and difference between the paired semantic representations of customer reviews and employee notes needs to be further extracted. Therefore, the element-wise product and subtraction on $(\bar{e}_i, \hat{e}_i)$ and $(\bar{c}_i, \hat{c}_i)$ (formulas (9–10)) are performed to obtain them [44,45], as these two operators can measure the correlation from two views of distance and angle, respectively. By applying formula (9), we assess the level of alignment between the pertinent details in customer reviews and relevant employee notes using $\bar{e}_i$ and $\hat{e}_i$, where e_{si} learns the comprehensive consistency relationship between $\bar{e}_i$ and $\hat{e}_i$ by element-wise product, and e_{di} captures the differences between $\bar{e}_i$ and $\hat{e}_i$ through subtraction. This enables us to detect customer requirements that are not comprehended by the employees. Likewise, by utilizing formula (10), we gauge the consistency between the relevant information in employee notes and customer reviews using $\bar{c}_i$ and $\hat{c}_i$. This allows us to identify customer demands that were misinterpreted by the employees. Furthermore, the consistency and related semantic features of customer reviews and employee notes can be obtained as follows $[e_{si}, c_{si}, e_{di}, c_{di}, \bar{e}_i, \bar{c}_i]$.

$$e_{si} = \bar{e}_i \odot \hat{e}_i, e_{di} = \bar{e}_i - \hat{e}_i \quad (9)$$

$$c_{si} = \bar{c}_i \odot \hat{c}_i, c_{di} = \bar{c}_i - \hat{c}_i \quad (10)$$

For controlling the scale of parameters, the two groups of features, $[\bar{e}_i, e_{si}, e_{di}]$ and $[\bar{c}_i, c_{si}, c_{di}]$, need to be further compressed. Specifically, a semantic compression layer is constructed to obtain low-dimension semantic vectors, calculated using the formulas (11, 12), where $\oplus$ stands for concatenation of vectors, W_e, W_c are semantic projection weights, and b_e, b_c are the bias of the projection.

$$t_e_i = ReLU(W_e[\bar{e}_i \oplus e_{si} \oplus e_{di}] + b_e) \quad (11)$$

$$t_c_i = ReLU(W_c[\bar{c}_i \oplus c_{si} \oplus c_{di}] + b_c) \quad (12)$$

Further, to compress and extract features at the sentence level, an isomorphic BiLSTM is used to uncover and integrate the semantic relationship of customer reviews and employee notes (formulas (13, 14)), where f_e_i and f_c_i capture the comprehensive semantic consistency, and h_{i-1}^f and h_{i-1}^b represent the forward and backward hidden states in the isomorphic BiLSTM, respectively.

$$f_e_i = BiLSTM(t_e_i, h_{i-1}^f, h_{i-1}^b) \quad (13)$$

$$f_c_i = BiLSTM(t_c_i, h_{i-1}^f, h_{i-1}^b) \quad (14)$$

3.4. Multi-Layer perception for learning NPS

Prior to engaging in the Multi-Layer Perception (MLP), the semantics relationship representations are aggregated by Meanpool token vectors so that the semantics and consistent degree in the sentence level can be effectively retained. Finally, these vectors are put into the final MLP to evaluate the NPS, as shown in the formula (15). In the training stage, the goal is to minimize the gap between predicted NPS and customer loyalty rating NPS_{real} , as shown in formula (16).

$$NPS = MLP\left(\sum_{i=1}^{l_e} \frac{f_e_i}{l_e}, \sum_{i=1}^{l_c} \frac{f_c_i}{l_c}\right) \quad (15)$$

$$MinL = \sum_{[c,e] \in dataset} [NPS - NPS_{real}]^2 \quad (16)$$

4. Experiments and evaluation

4.1. Pretest: semantic consistency inference and correlation verification

To verify whether the semantic consistency between customer reviews and employee notes for the same service impacts the final NPS, an analysis of the semantic consistency inference and correlation verification is conducted based on a pre-trained method and a customer service feedback dataset. The customer service feedback dataset contains 96,000 customer service entries derived from a world-class telecom company X in Europe. Each entry consists of the following three parts: customer review, employee note, and NPS rating. The range of the NPS is [0, 10]. Around 37.3 % of the NPS are in the range of [0, 6] (called ‘detractors’), 19.3 % are in the range of [7, 8] (called ‘passives’), and 43.4 % are in the range of [9, 10] (called ‘promoters’), following the telecom industrial conventions and NPS definition in [2]. The average length of customer reviews is 24.73 words, while the average length of employee notes is 30.86 words. It is also frequently observed that many low NPS entries (e.g., ≤ 6) are accompanied by short customer reviews (i.e., average 20.68 words), further emphasizing the necessity of using employee notes for in-depth NPS understanding and analysis.

Specifically, a pre-trained sequence inference model ESIM [44] is utilized to infer semantic relationship of customer reviews and employee notes. Moreover, a spearman correlation coefficient is used to detect the correlation between the semantic consistency and the NPS. The data and results of the pre-trained model, as well as the verification of the correlation between semantic consistency and NPS, are shown in Table 2.

Semantic consistency is defined as the degree of comprehensive consistency between customer reviews and employee notes, that is, emotional polarity, topic distribution and other potential semantic features. This degree variable can then be categorized into the following three types: consistent, partially consistent, and inconsistent, to analogize the three semantic relations in NLI. To detect the comprehensive semantic relationship, we use the ESIM model [44] trained using the SNLI dataset [34]. This is because ESIM utilizes chain modules to enhance the effect of sequence semantic inference, achieving an accuracy of 88.0 % for the SNLI dataset. Accordingly, we identified and labeled the semantic consistency between customer reviews and employee notes.

The SNLI dataset contained 570 K human-annotated sentence pairs, covering relations like entailment, neutral, and contradiction. This large-scale dataset is widely utilized and has been validated for its reliability in training NLI-related models to improve learning accuracy and label the semantic relationship between paired texts such as customer reviews and employee notes [34,35,47,48]. Furthermore, given that the semantic consistency labels and NPS are all ordered categorical variables, the spearman correlation coefficient $\rho =$

$\frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}$ is used to calculate the correlation between the semantic consistency label and the NPS.

In the customer service feedback dataset, the spearman correlation coefficient $\rho = 0.398$ with significance p-value = 0.0069. There is a significant correlation between semantic consistency and NPS, which lays the foundation for constructing an NPS understanding and analysis approach enhanced by the semantic consistency of customer reviews and employee notes. Here, the NPS can be explained from the perspective of the degree of fulfillment of customers’ needs, which is mirrored by the semantic consistency from customer reviews and employee notes.

4.2. Experimental setup

For preprocessing the unstructured raw data in customer service feedback dataset, word2vec is used to initialize words into the 128-dimension embedding vectors. Subsequent experiments from Section 4.2 onwards are conducted using the customer service feedback dataset for end-to-end model training and NPS analysis for all methods. The dimensions of BiLSTM, semantic compression layer, and Multi-Layer Perception are set as 64, 32, and 16, respectively. The parameters are learned by minimizing the Mean Squared Error of the training data with the initial learning rate = 0.001 and batch size = 32. The residual propagates backward in the hierarchical network and converges the model using an Adaptive gradient algorithm (Adagrad), which is effective for sparse data [49]. In the model training stage, batch normalization is deployed to further stabilize the training process, which greatly improves the training speed and suppresses the disappearance of the gradient. To avoid overfitting, the dropout rate is set as 0.25 for all feedforward connections [50].

4.3. Comparison with baseline methods

In the following experimental analysis, the widely used metrics, including MAE, RMSE, Hamming Distance, and Accuracy, are utilized to evaluate the learning performance of RNSC and baseline methods on the customer service feedback dataset. Generally, the smaller the MAE, RMSE, and Hamming Distance, the higher the performance, while the higher the Accuracy, the better the performance. Additionally, to justify the performance of the proposed RNSC approach, several important text-based rating analysis methods are treated as baseline methods for comparison.

CTR combines the merits of traditional collaborative filtering and probabilistic topic modeling, and develops an Expectation-Maximum algorithm to learn the maximum posteriori estimates (e.g., NPS) [22]. ALFM is an important aspect-aware potential factor model that can combine reviews and ratings for NPS evaluation [20], as it relaxes the constraint of one-to-one mappings between the latent topics and factors. NRT is a deep learning model that employs a multi-layer perceptron network to project customer and item vectors onto the label value through several layers of non-linear transformations and learn the customer rating ([51] August). Given that CTR, ALFM, and NRT methods originally only considered customer-generated text, employee notes are not considered by default. Therefore, the three methods are first conducted using data including only customer reviews. To further verify the value of including employee notes, the three methods are further adapted to deal with customer reviews and employee notes, that is, CTR with employee notes (CTR+e), ALFM with employee notes (ALFM+e), and NRT with employee notes (NRT+e). For RNSC, both customer reviews and employee notes are utilized.

The experimental results are shown in Table 3. First, the three baseline methods with employee notes (i.e., CTR+e, ALFM+e and NRT+e) perform better than those without (i.e., CTR, ALFM and NRT), with 15.39 %, 5.06 % and 5.20 % increase in Accuracy, respectively. This clearly verifies the value of including employee notes together with

Table 2
Semantic consistency inference and correlation verification.

Train	Data	SNLI dataset Label: entailment (2), neutral (1), contradiction (0) Size: 570,152 sentence pairs
	Model	ESIM [44]
	Optimization	Adam method [46]
	Accuracy	88.0 %
Test	Data	Customer service feedback dataset Label: consistent (2), partial consistent (1), inconsistent (0) Size: 96,000 customer review and employee note pairs
	Spearman correlation coefficient	0.398
	p-value	0.0069 **

Table 3
Comparison with baseline methods.

METHOD	Accuracy	Hamming Distance	MAE	RMSE
RNSC	68.87 %	0.3113	0.6473	1.0380
CTR	32.62 %	0.7673	1.2092	6.5802
ALFM	50.28 %	0.4972	0.7923	1.1729
NRT	58.31 %	0.4169	0.7669	1.2808
CTR+e	48.01 %	0.5200	0.7671	1.2378
ALFM+e	55.31 %	0.4469	0.8284	1.1116
NRT+e	63.51 %	0.3650	0.6507	1.1194

customer reviews. Second, the proposed RNSC performs the best out of all the baseline methods that treat the multi-source texts indiscriminately (i.e., a 5 % increase in Accuracy with respect to the second-best one, NRT+e). This shows that combining customer reviews and employee notes in terms of semantic consistency can help improve the performance of NPS learning. Although the baseline methods can be adopted to deal with multi-source texts, the indiscriminate treatment of ignoring the semantic consistency of the customer reviews and employee notes will inhibit learning performance.

4.4. Ablation study

To further validate the effectiveness of different components in the proposed RNSC approach, the following ablation analysis is conducted using the customer service feedback dataset to discard different components.

As presented in Table 4, when the customer reviews or employee notes are discarded, the learning performance significantly drops (e.g., a 17 % or 10 % drop on Accuracy). This demonstrates that customer reviews and employee notes contain key information for learning and understanding NPS. Customer reviews and employee notes are not simply duplicated information but their combination (i.e., the RNSC) remarkably improves the learning performance (i.e., 68.87 % on Accuracy). Second, if the attention mechanism is removed, the performance declines (e.g., from 68.87 % to 65.58 % on Accuracy). This demonstrates the importance of building a local consistency relationship based on the attention mechanism. Third, we explore the influence of the explicit design of semantic consistency recognition by moving element-wise product and subtraction components. If the relational operations are removed, the learning performance significantly drops (e.g., from 68.87 % to 60.92 % on Accuracy), highlighting the crucial role of capturing semantic consistency relationships in the NPS analysis context. Incorporating semantic consistency design can enhance the fitting accuracy of NPS, further substantiating the hypothesis that the semantic consistency of reviews and notes influences NPS analysis. In summary, the ablation study validates that the employee notes can effectively provide supplementary information about NPS and modeling from the perspective of the semantic consistency between the customer- and employee-generated content significantly enhances NPS learning and analysis.

4.5. Experiments on customer review sparse dataset

By combining customer- and employee-generated content, the data sparsity issue frequently observed in spontaneously filled customer reviews can be effectively counteracted. We conduct robustness

Table 4
Ablation experiments.

METHOD	Accuracy	Hamming Distance	MAE	RMSE
RNSC	68.87 %	0.3113	0.6473	1.0380
RNSC /review	51.91 %	0.4809	1.0329	2.6848
RNSC /note	58.07 %	0.4193	0.8644	1.4955
RNSC /attention	65.58 %	0.3319	0.6861	1.1226
RNSC /consistency	60.92 %	0.3707	0.7745	1.5374

experiments on the dataset with sparse customer reviews to prove that. We concretely extract entries with short review (i.e., length of words in it less than 6) from the customer service feedback dataset, which provides a total of 37,980 pieces of data (39.6 % in the total). Fig. 3 shows the experimental results for the sparse dataset compared with the results of the original dataset. First, the performance of all methods, including RNSC, decreases for the sparse dataset (i.e., decrease on Accuracy, increase on Hamming distance, MAE, and RMSE), showing that sparsity weakens the learning effect. Nevertheless, it is observed that the methods incorporating employee notes (i.e., CTR+e, ALFM+e, and NRT+e) are advantageous because they counteract sparsity in customer reviews; therefore, this method is better than the other methods without employ notes (i.e., CTR, ALFM, and NRT). Concretely, the Accuracy of CTR+e, ALFM+e, and NRT+e is reduced by 11.31 %, 16.74 %, and 20.18 %, respectively, in the sparse case, while CTR, ALFM, and NRT are reduced by 28.92 %, 32.31 % and 29.47 % on Accuracy, respectively. This further confirms that integrating internal and external multi-source data, such as employee notes and customer reviews, can largely mitigate the data sparsity issue frequently observed across external customer-generated data.

Additionally, RNSC shows the best performance in the sparsity dataset compared to the other baseline methods (e.g., 62.20 % on Accuracy). Moreover, the Accuracy of RNSC is only reduced by 6.67 %, compared with the original case, which also shows the smallest performance drop compared with all the baseline methods. Similar out-performance of RNSC while tolerating sparsity can be found for Hamming distance, MAE, and RMSE results. These results further highlight that the RNSC approach enhanced using semantic consistency can greatly benefit from the paired information, leading to high robustness while dealing with sparsity.

4.6. Diagnostic analysis of NPS with granular words

Based on the trained RNSC approach, the attention mechanism is integrated to calculate the contribution of the words in customer reviews and employee notes to the learned representation of semantic consistency e_{si} , c_{si} and e_{di} , c_{di} . This is because the attention mechanism is widely recognized for helping to realize the fine-grained tracing of potential information association [52]. Specifically, we calculate attention scores of words in customer reviews and employee notes to semantic consistency representations using the formulas (6, 7).

Therefore, the diagnostic analysis of NPS, particularly for low scores, can be skillfully conducted by incorporating semantic consistency and attention mechanism. One of major benefits of RNSC approach is that it allows us to gain a comprehensive understanding of NPS by examining the influence of inconsistent evaluations of customer demands within customer reviews and employee notes. These inconsistencies are identified through semantic consistency scores. Additionally, the utilization of attention mechanism as well as scores helps us pinpoint specific words, providing precise insights to comprehend semantic consistency and shed light on crucial areas within customer reviews and employee notes. For illustrative purposes, some representative samples are discussed, and their diagnostic results are listed in following Table 5. In the table, the words with high attention weights are highlighted.

Table 5 reveals that the employees tend to provide more details regarding the operational purpose and potential exemption. This is especially true for low NPS customers (i.e., detractors), which does provide additional and valuable information for in-depth NPS understanding and diagnosis. In the first sample, the highlighted words include “powerline,” “couldn’t,” “replacement,” “long,” etc. They indicate the key crux for why NPS = 6. The customer complains that the powerline is not working, while the employee mentions the problems are the inability to change the order and the long waiting time, which shows partial semantic inconsistency between reviews and notes leads to an NPS = 6. In the second sample, the selected words from employee notes, which include “issues,” “have,” “to find,” and “happy,” report the

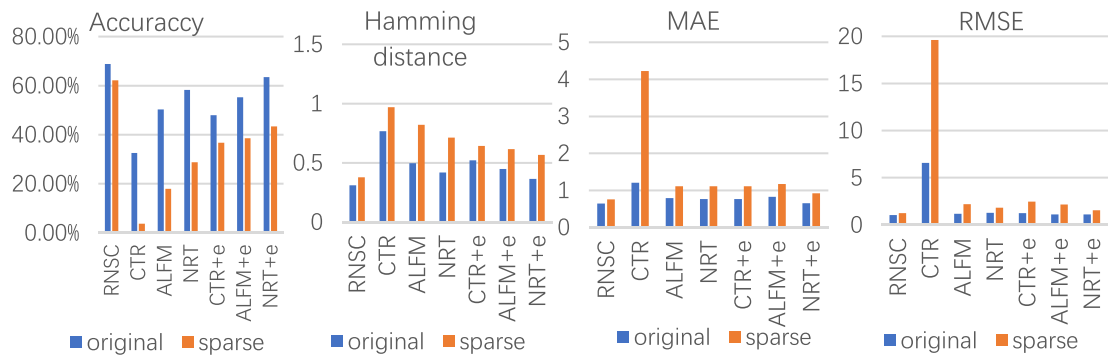


Fig. 3. Experimental results of original and sparse dataset.

Table 5
Diagnostic results of some samples with low NPS.

No.	Customer Review and Employee Note	NPS
1	<u>Employee:</u> called to inform that one of the powerline adapters is not working. Couldn't place a replacement order on the same. Place an order for mini connectors and also asked the advisor to waive off the charges as customer is on contract. Order placed successfully. <u>Customer:</u> Because I was on the phone 38 min just for a faulty item much too long.	6
2	<u>Employee:</u> Customer is having issues with broadband connection and is unable to connect to the internet. I have run a test flow on consumer on landline and also bb connection. I have not managed to find a fault, so I have carried a home check which also has not showed me anything. I have now arranged an engineer visit to solve the problem. Customer happy with result and happy to close the complaint. <u>Customer:</u> The line was interrupted, he promised me to phone back and did not, broadband is still crap.	0
3	<u>Employee:</u> customer not informed calls going over 60 min will be charged. Complaint raised at customer request, and the cost of the call refunded as goodwill. <u>Customer:</u> I do not trust (the company) as a supplier. I am forced to have a landline I don't want and then tricked into extra charges. However, he was professional and dealt politely and efficiently with my complaint.	0
4	<u>Employee:</u> customer called regarding the engineer visit charges of £129.99, had a word with EHIS team and raised a request to investigate regarding the charges and informed the customer that she will getting a call from case manager in next 5 working days. <u>Customer:</u> Appalling service I don't expect you to take money from my account when it was you people that messed up and I expect to have to wait to get it returned I am furious with you.	0
5	<u>Employee:</u> Customer is having issues with broadband connection and is unable to connect to the internet. I have run a test flow on consumer on landline and also connection, I have not managed to find a fault so I have carried a home check which also has not showed me anything. I have now arranged an engineer visit to solve the problem. <u>Customer:</u> The line was interrupted, he promised me to phone back and did not, broadband is still bad.	0
6	<u>Employee:</u> walked her through forgetting the networks left her only with hub on the MacBook, no security message anymore NFAR. <u>Customer:</u> Happy with the way the person handled my call. Very unsatisfied with BT in general, email account hacked ongoing issues suspicious emails all repo. <u>Customer:</u> She was helpful although BT couldn't provide me with the info I needed.	4

product faults, the enterprise’s response, and self-evaluated attitude from the employee’s perspective. Moreover, the selected words “phone,” “back,” and “line” in customer reviews reflect the service inefficiency and unresolved faults perceived by the customer. This sample explicitly reveals the key problems and strong semantic inconsistency, which clearly reflect why NPS = 0. A similar diagnostic analysis could also be conducted for the other samples. Overall, the proposed RNSC approach, combining customer reviews and employee notes through semantic consistency, can derive NPS with satisfactory learning performance and

facilitate diagnostic analysis in a granular manner, enhancing in-depth NPS understanding and analysis.

5. Conclusion

As the key indicator for measuring customer loyalty, NPS has attracted significant attention across academia and industry. In NPS understanding and analysis, few studies have considered multi-source data, such as customer- and employee-generated content. Most studies have conducted analyses using only customer reviews, which are inevitably affected by weak information. In this study, we introduce information from employee notes that are from internal enterprise and could effectively complement external customer-generated data. This also provides a new exploration direction for in-depth NPS diagnosis from the perspective of integrating internal and external generated information.

Moreover, considering the semantic relationship between these two information sources and the customer service context, semantic consistency is defined to evaluate the semantic relationship between customer reviews and employee notes. These aspects, to a large extent, can reflect the degree of the fulfillment of customers’ needs to better understand and analyze NPS. Based on these, a hierarchical deep learning approach, namely RNSC, is proposed to enhance semantic consistency recognition. It can effectively capture the important semantic consistency relationship to enhance NPS analysis by learning from the paired texts and achieve satisfactory learning performance. Extensive data experiments, such as comparison with baseline methods, ablation experiments, and sparse data experiments, verified the out-performance of the proposed RNSC approach. Additionally, the integrated attention mechanism could also detect and output the keywords that significantly contributed to the semantic consistency, showing its potential for supporting diagnostic analysis and furthering service optimization.

Overall, the major contribution of the study is threefold. First, a novel NPS analysis framework is designed to effectively manage multi-resource texts from internal and external enterprises, that is, customer reviews and employee notes. These aspects provide ample knowledge about NPS and eliminate the problem of weak external customer review. Second, semantic consistency is designed and measured using paired content generated by customers and employees. Third, the proposed deep learning approach RNSC enhanced using semantic consistency outperforms in the learning performance and analytical effect.

Our study also has certain limitations. First, we did not have data on customer profiles and service logs, which prevented us from linking the NPS with more fruitful related information. Second, the theory driving customer loyalty needs to be further investigated, and the internal mechanism of semantic consistency should be carefully addressed to strengthen theory-building. Third, the experimental analysis was conducted in a telecom company context. Data experiments from other contexts should be considered in future studies to validate the

generalizability of the proposed framework and approach. Fourth, from a purely technical perspective, the proposed RNSC approach can be further improved in structure optimization, parameter tuning, and more. Additionally, more comparable deep-learning methods could be involved in future data experiments.

CRedit authorship contribution statement

Xiaowei Shi: Writing – original draft, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Qiang Wei:** Writing – review & editing, Validation, Supervision, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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